



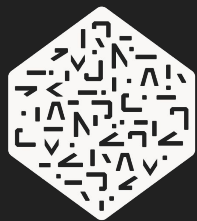
Introduction to Numerai

By: Aventurine

Overview

Numerai is a data science competition where you build machine learning models to predict the stock market. You are provided with free, high quality data that you can use to train models and submit predictions daily.

Numerai computes the performance of these predictions over the following month and you can stake NMR on your model, to earn (or burn) based on your model's performance.



NUMERAI

Data

- Numerai's free dataset is made of clean and regularized financial data.
- The dataset is **obfuscated** so that it can be given out for free and modeled without any financial domain knowledge.
- This also means that models you build on this data cannot be used outside of the Numerai tournament.

id	era	feature1	...	feature310	target
n2b2e3dd163cb422	era1	0.75	...	0.00	0.25
n177021a571c94c8	era1	1.00	...	0.25	0.75
n7830fa4c0cd8466	era1	0.25	...	1.00	0.00
nc584a184cee941b	era1	0.25	...	0.00	1.00
nc5ab8667901946a	era1	0.75	...	0.25	0.25
n84e624e4714a7ca	era1	0.00	...	0.75	1.00

Numerai's obfuscated dataset

Data

Here is an example of how to get our dataset:

```
from numerapi import NumerAPI
import pandas as pd

napi = NumerAPI()
napi.download_dataset("v5.0/train.parquet")
training_data = pd.read_parquet("v5.0/train.parquet")
```

Each row in the dataset corresponds to a specific stock at a specific point in time. The point in time is noted by the era (This is the only grouping and time series information the data exposes)- each represents a week. The IDs are unique in each era such that you cannot match stocks across eras - this is necessary for the obfuscation. The features are quantitative attributes known about the stock at the time (e.g P/E ratio, ADV, etc.). The target is a measure of stock market returns 20 days into the future where low means bad performance and high means good performance.

Data

Versions

There are several versions in the data API because Numerai is constantly improving their dataset. In general, if you are building a new model you are encouraged to use the latest version. Each minor version (i.e. v4 vs v4.1 vs v4.2) generally maintains backwards compatibility with each other and makes it easy for you to plug in the latest data to your trained models. Major versions (i.e. v3 vs v4) may have large, breaking changes to the structure and/or contents of the datasets, so it's usually best to re-train your models when major versions are released.

Formats

The main file format of the Numerai data API is Parquet, which works great for large columnar data - I highly encourage that you become familiar with Parquet if you are not already as it is a widely-used format in big data and data science.

Modeling

- Your objective is to build machine learning models to predict the target given the features. You can use any language or framework that you like.
- Here is an example model in Python using LightGBM, but you can use any language or framework that you like.

```
import lightgbm as lgb

features = [f for f in training_data.columns if "feature" in f]

model = lgb.LGBMRegressor(
    n_estimators=2000,
    learning_rate=0.01,
    max_depth=5,
    num_leaves=2 ** 5,
    colsample_bytree=0.1
)
model.fit(
    training_data[features],
    training_data["target"]
)
```

Submissions

- Each day (Tuesday through Saturday), new live data is released. This represents the current state of the stock market. You must generate live predictions and submit them to Numerai. You are asked to submit a prediction value for each id in the live data.
- Behind the scenes, Numerai combines the predictions of all models into the *Stake-Weighted Meta Model*, which in turn is fed into the Numerai Hedge Fund for trading.

This is what a submission might look like:

id	prediction
n60dffdiceb7e467	0.25
nadaeef0214b84a8	1.00
nb13883520a4344f	0.25
n423766c5a4fa42a	0.75
n252b14301e46a31	0.25
n75a5baf93a624cc	0.00
n2ff91086716e413	1.00

A sample submission

Submissions

Cloudpickle

Pickle your function with the [cloudpickle](#) library and upload the pickle file to Numerai:

- This library is used in other computing frameworks like [Dask](#) and [Ray](#) and allows you to easily run your local Python code remotely in the cloud. The main benefit of using cloudpickle over the standard [pickle](#) library is that it serializes your local context along with your code. This makes it very convenient to package up code developed locally in any environment.
- In the example above, our predict function references model and feature_cols defined in the global scope. Cloudpickle is smart enough to correctly serialize both model and feature_cols by value so that it is also available when this function is run by Numerai.

```
# Use the cloudpickle library to serialize your function
import cloudpickle
p = cloudpickle.dumps(predict)
with open("predict.pkl", "wb") as f:
    f.write(p)
```

Copy

Scoring

Submissions are scored against two main metrics:

Correlation (CORR): Your prediction's correlation to the target

Meta Model Contribution (MMC): Your prediction's contribution to the Meta Model

Since the target is a measure of 20 business days of stock market returns, it takes about 1 month for each submission to be fully scored.

Scoring

How is CORR computed?

- First the predictions are gauss ranked. Numerai does this to match our live trading process where all model predictions are standardized this way before being ensembled together in the meta model.
- Then the target is centered around 0. Numerai this to match the gauss ranked predictions which are now centered around 0.
- Finally both the gauss ranked predictions and the centered target are raised to the power of 1.5 before calculating the Pearson correlation. Numerai does this to accentuate the tails as hedge fund tends to only trade the stocks with highest or lowest predicted returns.

```
def numerai_corr(preds, target):  
    # rank (keeping ties) then Gaussianize predictions to standardize prediction distribution  
    ranked_preds = (preds.rank(method="average").values - 0.5) / preds.count()  
    gauss_ranked_preds = stats.norm.ppf(ranked_preds)  
  
    # make targets centered around 0  
    centered_target = target - target.mean()  
  
    # raise both preds and target to the power of 1.5 to accentuate the tails  
    preds_p15 = np.sign(gauss_ranked_preds) * np.abs(gauss_ranked_preds) ** 1.5  
    target_p15 = np.sign(centered_target) * np.abs(centered_target) ** 1.5  
  
    # finally return the Pearson correlation  
    return np.corrcoef(preds_p15, target_p15)[0, 1]
```

The key takeaway here is that your prediction's distribution does not matter. You are only evaluated on your prediction's ranks. And your score depends more on the tails than a typical rank-correlation.

Scoring

What is MMC (and BMC)?

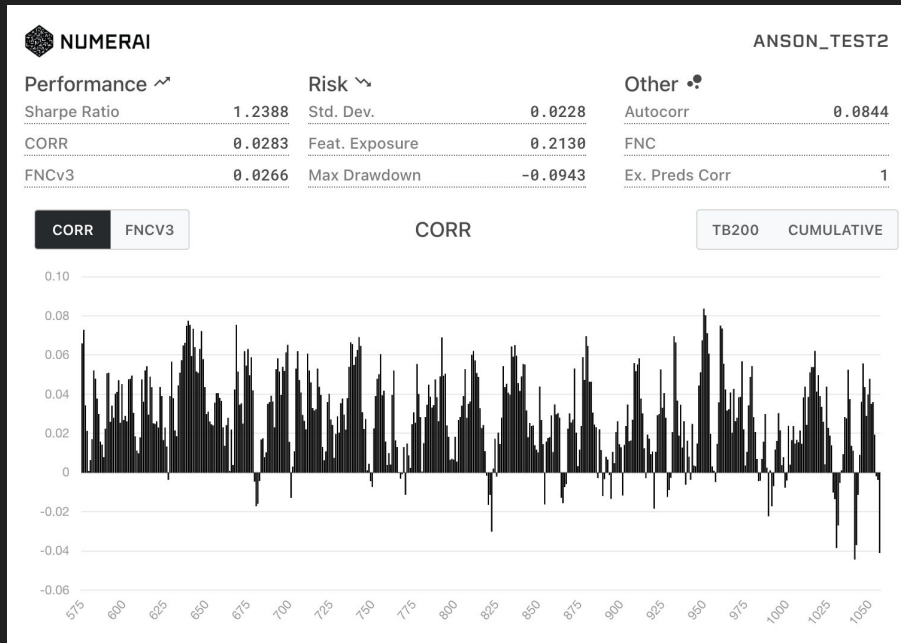
- Meta Model Contribution (MMC) is the covariance of a model with the target, after its predictions have been neutralized to the Meta Model. Similarly, Benchmark Model Contribution (BMC) is the covariance of a model with the target, after its predictions have been neutralized to the stake-weighted Benchmark Models.
- These metrics tell us how the unique component of a model contributes to the correlation of the Meta Model (or the Benchmark Models in the case of BMC). By neutralizing the model's predictions by the Meta Model or Benchmark Models, the remaining orthogonal component's covariance with the target is that model's contribution.

Scoring

Scores not used for payouts

- Feature Neutral Correlation (FNC): Your prediction's correlation to the target after neutralizing against all features
- Correlation with the Meta Model (CWMM): Your prediction's correlation to the Meta Model (stake weighted average of all predictions).
- Benchmark Model Contribution (BMC): Your prediction's correlation to the target after neutralizing against the stake-weighted Benchmark Models.

Diagnostics



- The diagnostics tool computes and charts your scores over the validation dataset.
- If you uploaded your model via [Model Upload](#), then Numerai will automatically run your model over the validation dataset to generate diagnostics. If you wish, you may also manually run diagnostics by heading over to [numer.ai/scores](#) and clicking on the Run Diagnostics button:

“Steaking”

When you are ready and confident in your model's performance, you may stake on it with NMR - Numerai's cryptocurrency. After the 20 days of scoring for each submission, models with positive scores are rewarded with more NMR, while those with negative scores have a portion of their staked NMR *burned* (destroyed such that no one, not even Numerai, can access it).

Staking serves two important functions:

- 1) "Skin in the game" allows Numerai to trust the quality of staked predictions.
- 2) Payouts and burns continuously improve the weights of the Meta Model.

“Steaking”

Numeraire (NMR)

NMR is the cryptocurrency that powers staking and payouts on Numerai.

To stake NMR on your model you must first acquire NMR, then deposit it into your [Numerai wallet](#). When you are ready, you can withdraw it to a given eth address.

Depositing NMR

- [Coinbase](#) - a safe and easy place to buy NMR with USD, GBP, EUR or Bitcoin. A good place to start if you are new to crypto. Availability depends on your region.
- [Uniswap](#) - a decentralized exchange you can use to swap Ethereum based tokens like ETH or USDC for NMR. Requires you to bring your own wallet like MetaMask. Great for DeFi enthusiasts.

How to stake

The screenshot shows the 'Stake' tab of the Numer.ai staking interface. At the top, there are three tabs: 'Stake' (selected), 'Release', and 'Advanced'. Below the tabs, the 'Stake Amount' is displayed as '0.1' with a blue 'NMR' label to its right. Below the input field, the available balance is shown as 'Available: 0.450808483525612275 NMR (max)'. At the bottom, there is a large black button labeled 'Confirm'.

- Head over to numer.ai/staking, and click on the "manage stake" button to open the stake modal.
- Use the Stake form to stake NMR from your wallet on your model.
- Staked NMR will remain locked until you release it back to your wallet, which takes 1 month. Specifically, the NMR will be released at the resolution of your last active round at the time of the request. While pending release, the unstaked amount may still be subject to burns but will not count towards upcoming payouts.
- You can optionally configure where your earnings go and the score multipliers used to compute your payouts.

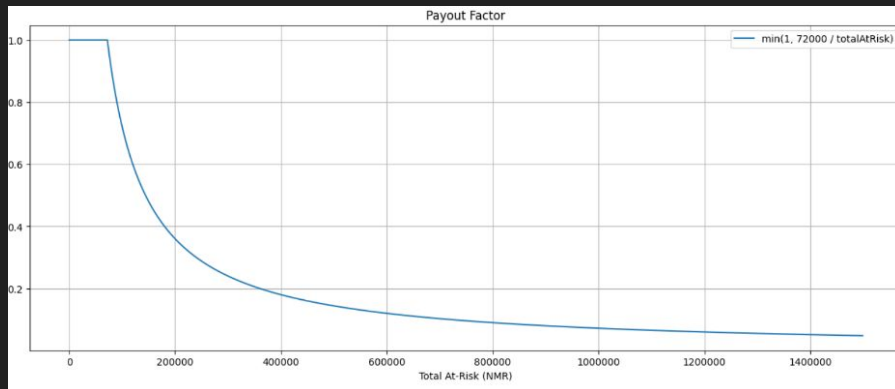
Payouts

Your payout is primarily a function of your scores. If you have a positive score you will get a payout. If you have a negative score a portion of your stake will burn.

The maximum payout or burn per round is capped at $\pm 5\%$

Stake is your model's stake value at the close of the round. This is also referred to as the stake value at-risk for a round. Your stake value at-risk for a round does not include any unstaked amounts that are pending release, and is set to 0 if you have no valid submission for a round.

`payout_factor` is a dynamic value that scales inversely with total NMR staked based on the `staking_threshold`.



The Leaderboard

STAKED MODELS

5,561

TOTAL NMR STAKED

1,055,844 NMR

AVG 1 YEAR RETURN

+62.07%

LEADERBOARD SEARCH

		CDR2D	CDR2Dv2	CDR2D0	PRCV3	▼ TC	STAKE	1 DAY	3 MONTH	1 YEAR
1	 SHATTERED3xTC	0.0246	0.0152	0.0419	0.0193	0.0365	23,497.24 NMR	3.2%	26.2%	615.9%
2	 K3_D4 3xTC	0.0068	0.0071	0.0093	0.0091	0.0343	672.86 NMR	2.9%	—	—
3	 SHATT032 3xTC	0.0237	0.0127	0.0410	0.0186	0.0312	0.041 NMR	6.3%	22.7%	—
4	 ESG_18 3xTC	0.0189	0.0106	0.0363	0.0099	0.0309	0.027 NMR	4.8%	77.5%	—
5	 ML_IG_LYP	0.0305	0.0221	0.0474	0.0159	0.0302	0.15 NMR	2.5%	16.6%	63.9%
6	 ESG_2 3xTC	0.0138	0.0095	0.0258	0.0080	0.0300	97.97 NMR	5.9%	73.3%	—
7	 SANFRANCISCO 3xTC	0.0182	0.0137	0.0342	0.0079	0.0281	0.070 NMR	3.9%	10.3%	49.9%
8	 HYDRASIRAFFE 3xTC	0.0210	0.0157	0.0336	0.0149	0.0280	26.3 NMR	9.2%	91.1%	133.5%
9	 SHATT021 3xTC	0.0264	0.0154	0.0413	0.0223	0.0269	0.041 NMR	0.40%	21.3%	—
10	 REPUTATION 3xTC	0.0170	0.0095	0.0312	0.0069	0.0267	31,445.72 NMR	4.0%	62.7%	167.8%
11	 ESG_6 3xTC	0.0144	0.0086	0.0266	0.0061	0.0264	69.29 NMR	6.1%	26.9%	—
12	 ESG_20 3xTC	0.0188	0.0085	0.0367	0.0085	0.0263	0.021 NMR	5.9%	64.6%	—
13	 GOLDMH_6	0.0202	0.0123	0.0401	0.0134	0.0263	—	—	—	—
14	 PRESIDIO 3xTC	0.0202	0.0151	0.0398	0.0107	0.0260	1.6 NMR	4.3%	11.4%	52.5%
15	 SPOTTY_2 3xTC	0.0185	0.0118	0.0254	0.0124	0.0257	1.34 NMR	5.3%	56.8%	—

numer.ai/leaderboard

- Only the final scores for rounds count towards a model's live performance.
- The 1 year average score is also called reputation and your model's rank on the leaderboard is primarily based on your model's 1 year average TC score.

Other Numerai Tournaments

Numerai Signals Introduction

While the [Numerai Tournament](#) provides free data, [Numerai Signals](#) asks you to bring your own data - your own unique signal. A signal in the stock market is a feed of information. It is numerical data about stocks that can be used by quantitative hedge funds like Numerai. These feeds of information are used to model the stock market and construct portfolios (collections of stock holdings).

Numerai Crypto Introduction

Similar to [Numerai Signals](#), [Numerai Crypto](#) asks you to bring your own data - your own unique signal. A signal in the crypto market (or "CryptoSignal") is a feed of information. It is numerical data about tokens that can be used by crypto traders. These feeds of information are used to model the crypto market and construct the Numerai Crypto Meta Model, which Numerai gives back to you **free of charge**.

Thank You!

Twitter/Telegram: [aventurine_eth](#)

Email: aventurine.eth@proton.me

